INSTRUCTOR MANUAL

CSE 4571 — Practical Robotics Project with Arduino

Seventh Semester B.Tech (2L – 4P) | Hardware-Integrated Simulation-Based Laboratory

Practical Arduino Robotics — Lukas Kaul

SESSION-2

Analog Sensor Interfacing and Real-Time Data Acquisition

Learning Domain Distribution — Cognitive: 45% | Psychomotor: 40% | Affective: 15%

*Note: This is a **Hardware-Integrated Simulation-Based Laboratory** course, where simulation activities are supported by actual hardware components. **Physical components, breadboards, and hardware kits are required** for laboratory activities, along with the prescribed simulation tools.*



SIKSHA 'O' ANUSANDHAN

1. Session Information

Field	Details
Course Code / Title	CSE 4571 — Practical Robotics Project with Arduino
Semester	Seventh Semester B.Tech (CSE)
Course Format	2L – 4P (Practical / Laboratory, Hardware-Integrated and Simulation-Based)
Session	2
Duration	6 Hours (360 minutes)
Software Used	Tinkercad/ Wokwi (Online Circuit Simulation Platform)
Topic	Analog Sensor Interfacing and Real-Time Data Acquisition
Textbook Reference	Practical Arduino Robotics — Lukas Kaul

*Hardware components, apparatus, component kits, and breadboards are integrated where required, while simulation-based activities are performed on-screen using **Tinkercad/Wokwi**.*

2. Learning Objectives

By the end of this session, students will be able to:

Objectives	Statements
O1	Explain the fundamental concept of analog sensors and their role in measuring physical parameters for robotic and embedded-system applications.
O2	Differentiate between analog and digital sensor outputs, and identify the major terminals, signal characteristics, and interfacing requirements of a typical analog sensor.
O3	Explain how an analog sensor converts a physical parameter into an electrical voltage and describe how the Arduino UNO ADC converts the sensor voltage into a corresponding digital value.
O4	Identify and correctly wire the Arduino UNO and an analog sensor in a simulation or hardware setup, including signal, power supply, and common-ground connections.
O5	Implement and observe Arduino-based real-time analog data acquisition by reading sensor values through the ADC, displaying the acquired data through the Serial Monitor/Serial Plotter, and relating the measured values to practical robotic applications.

3. Course Outcomes (CO1–CO6)

CO	Course Outcome Statement	Learning Domain	Taxonomy Level
CO1	Explain the fundamentals of Arduino-based robotic systems and interface different sensors for measurement, analog data acquisition, and real-time sensing applications.	Cognitive (L) + Psychomotor (P) + Affective (A)	L2 – Understand+ P1: Perception+ A1: Receiving

CO2	Develop Arduino programs to control robotic actuators including servo motors, stepper motors, and DC motors for position and motion-based applications.	Cognitive (L) + Psychomotor (P) + Affective (A)	L3 – Apply + P3 – Guided Response+ A2: Responding
CO3	Implement intelligent robotic control systems using sensor inputs, finite state machines, non-blocking programming techniques, and feedback-based control mechanisms.	Cognitive (L) + Psychomotor (P) + Affective (A)	L3 – Apply + P4 – Mechanism + A3: Valuing
CO4	Design modular and reusable robotic software components using object-oriented programming concepts and Arduino libraries for embedded applications.	Cognitive (L) + Psychomotor (P) + Affective (A)	L5 – Evaluate + P5 – Complex Overt Response+ A4: Organization
CO5	Analyze and troubleshoot robotic systems by integrating communication interfaces, remote-control techniques, and wireless monitoring using embedded platforms.	Cognitive (L) + Psychomotor (P) + Affective (A)	L4 – Analyze + P6 – Adaptation + A4: Organization
CO6	Design and demonstrate an Arduino-based robotic application by integrating sensors, actuators, control algorithms, and wireless communication modules while following engineering practices.	Cognitive (L) + Psychomotor (P) + Affective (A)	L6 – Create + P7 – Origination + A5 – Characterization

L: Cognitive (L1: Remembering; L2: Understanding; L3: Applying; L4: Analyzing; L5: Evaluating; L6: Creating)

P: Psychomotor (P1: Perception; P2: Set; P3: Guided Response; P4: Mechanism; P5: Complex Overt Response; P6: Adaptation; P7: Origination)

A: Affective (A1: Receiving; A2: Responding; A3: Valuing; A4: Organization; A5: Characterization)

The course CSE 4571 - Practical Robotics Project with Arduino primarily reflects the following Sustainable Development Goals (SDGs):

UN Sustainable Development Goals (SDGs)

SDG	Goal	Course Relevance
SDG 4	Quality Education	Develops practical skills in analog sensor interfacing, Arduino programming, ADC-based data acquisition, real-time monitoring, circuit implementation, simulation, observation, and troubleshooting.
SDG 7	Affordable and Clean Energy	Promotes awareness of efficient electronic system operation and responsible energy use through sensor-based monitoring, appropriate signal processing, and controlled operation of Arduino-based systems.
SDG 9	Industry, Innovation and Infrastructure	Promotes innovation in Arduino-based sensing and data acquisition systems through analog sensor interfacing, real-time data monitoring, simulation/hardware implementation, and applications in smart systems, automation, robotics, and IoT.
SDG 12	Responsible Consumption and Production	Encourages responsible engineering practices through simulation-based testing, systematic troubleshooting, efficient use of electronic components, proper sensor interfacing, and software-based monitoring to reduce unnecessary component usage and potential damage.

CO – PO Mapping (Focus: CO1-CO6)

CO	PO Mapped	Strength
CO1	PO1, PO2	High
CO2	PO1, PO5	High
CO3	PO1, PO4, PO5	High
CO4	PO2, PO4, PO5	High
CO5	PO3, PO5, PO6	High
CO6	PO3, PO5, PO9, PO10, PO11	High

4. Learning Domain Distribution (3H Model)

The 360-minute session strictly maintains the prescribed course-level domain distribution of Cognitive 40%, Psychomotor 45%, and Affective 15%.

Domain	Weightage	Time (approx.)	Focus
Cognitive (Head)	45%	162 minutes	Understanding analog sensor operation, physical parameter measurement, analog and digital signals, sensor output voltage, Arduino ADC, ADC-to-digital value conversion, real-time data acquisition, Arduino programming, sensor calibration, and interpretation of acquired sensor data.
Psychomotor (Hand)	40%	144 minutes	Identifying and wiring the Arduino UNO and analog sensor; making signal, power, and common-ground connections; running the Arduino program; acquiring analog sensor readings; observing real-time values through the Serial Monitor/Serial Plotter; performing simulation/hardware testing; and troubleshooting sensor interfacing and data acquisition.
Affective (Heart)	15%	54 minutes	Developing safety awareness, careful wiring, systematic observation, responsible handling of electronic components, and accurate recording of sensor data, teamwork, recognition of sensor operating limitations, and safe and responsible operation of the Arduino-based data acquisition system.

Bloom's Taxonomy Mapping

Bloom's Level	Activity
Remember	Recall the names and functions of the Arduino UNO, analog sensor, sensor terminals, power supply, GND, and analog input pins used for sensor interfacing.
Understand	Explain the working principle of an analog sensor, the relationship between a physical parameter and sensor output voltage, and the role of the Arduino ADC in converting the analog signal into a digital value.
Apply	Build or simulate the analog sensor–Arduino circuit and correctly connect the VCC, GND, signal, and analog input terminals for real-time sensor data acquisition.
Analyze	Identify sensor terminals, verify correct wiring and signal flow, and analyze the acquired sensor data by relating ADC readings to changes in the measured physical parameter through the Serial Monitor/Serial Plotter.

5. Instructor Preparation

Before opening Tinkercad/Wokwi, the instructor should explain the basic concepts of analog sensors, voltage-divider circuits, Arduino analog input pins, ADC operation, and real-time data acquisition. Introduce the role of sensors in robotic and automation systems, where physical quantities such as light, temperature, pressure, or distance are converted into electrical signals that can be measured by a microcontroller.

Briefly explain the difference between **analog and digital signals**, emphasizing that analog sensors produce continuously varying voltage signals, while digital sensors generally provide discrete logic levels or digital data. Demonstrate the relationship between the sensor output voltage and the physical quantity being measured. Explain that the Arduino UNO contains a 10-bit Analog-to-Digital Converter (ADC), which converts an input voltage between 0 and 5 V into a numerical value from 0 to 1023.

The instructor should demonstrate how an analog sensor can be connected to an Arduino analog input and how the acquired data can be displayed in real time through the Serial Monitor or Serial Plotter. The experiment provides a foundation for sensor-based robotic systems, data monitoring, feedback control, and automation applications.

Analog Sensor Interfacing and Real-Time Data Acquisition

An analog sensor is a device that produces a continuously varying electrical output corresponding to a physical quantity. In this experiment, a **Light Dependent Resistor (LDR)** is used as the analog sensing element. The resistance of the LDR changes according to the intensity of incident light. When combined with a fixed resistor in a voltage-divider circuit, this change in resistance produces a varying voltage that can be measured by the Arduino UNO.

The Arduino reads this analog voltage through one of its analog input pins. The built-in ADC converts the measured voltage into a corresponding digital value. The acquired value can then be transmitted to a computer through the USB serial connection, allowing real-time observation of the sensor response.

The experiment demonstrates the complete sequence of analog data acquisition:
Physical Quantity → Sensor → Voltage Signal → Arduino ADC → Digital Value → Serial Monitor
The process is useful for applications such as automatic lighting, environmental monitoring, industrial automation, and sensor-based robotic systems.

Analog Sensor Data Acquisition Flow

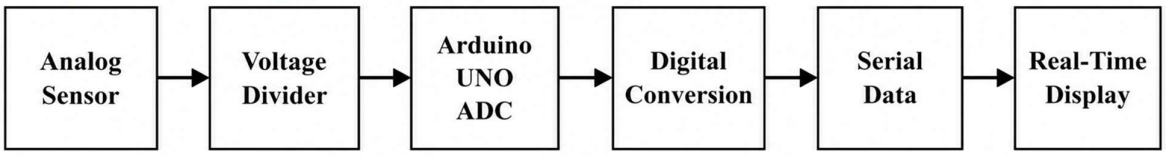


***Figure 1**. Functional flow of analog sensor interfacing and real-time data acquisition.*

Theory: Analog Sensor as a Data Source

An analog sensor generates an electrical signal that varies continuously with the physical quantity being measured. Unlike a digital sensor, which generally provides discrete logic states or digitally encoded information, an analog sensor can produce many intermediate voltage levels.

In this experiment, an LDR is used to detect changes in light intensity. The resistance of the LDR decreases when the intensity of light increases and increases when the light intensity decreases. The LDR is connected with a fixed resistor to form a voltage-divider circuit.

The output voltage from the voltage divider is connected to the Arduino UNO analog input pin A0. The voltage measured at A0 varies according to the resistance of the LDR and, therefore, according to the surrounding light intensity.

For a voltage-divider circuit:

$$\mathbf{Vout = Vin \times R2 / (R1 + R2)}$$

where:

- **Vin** = Arduino supply voltage
- **Vout** = voltage applied to the analog input
- **R1** = LDR resistance
- **R2** = fixed resistor

The Arduino UNO converts the analog input voltage into a 10-bit digital value. Since a 10-bit ADC provides:

$$\mathbf{2^{10} = 1024\ levels,}$$ the resulting ADC value ranges from: **0 to 1023.**

For a 5 V reference voltage, the approximate input voltage can be calculated as:

$$\mathbf{Voltage = ADC\ Value \times 5.0 / 1023}$$

Thus, the Arduino can continuously measure the sensor output and transmit the acquired information to a computer for real-time monitoring.

Circuit Identification and Wiring

Students should identify the **LDR, fixed resistor, Arduino UNO, breadboard, and jumper wires** before wiring.

The LDR and fixed resistor are connected as a voltage-divider circuit. The junction between the LDR and resistor is connected to **A0** of the Arduino UNO. The circuit is powered using the Arduino **5 V and GND** pins. All grounds are common.

Component	Connection
LDR terminal 1	5V
LDR terminal 2	A0
10 kΩ resistor terminal 1	A0
10 kΩ resistor terminal 2	GND
Arduino 5V	Sensor circuit supply
Arduino GND	Common ground

Wiring Connections:

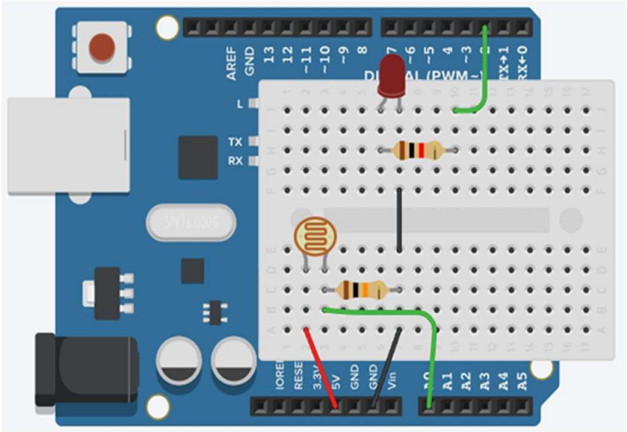


Figure 2. Tinkercad/Wokwi circuit setup for for analog sensor interfacing and real-time data acquisition.

Figure 3. Hardware setup for analog sensor interfacing and real-time data acquisition.

Theory: ADC and Real-Time Data Acquisition Concept

The Arduino UNO uses a **10-bit ADC** for converting the analog input signal into a digital value. The analog input pins can measure voltages within the configured ADC reference range. With the default 5 V reference, an input of approximately 0 V corresponds to an ADC value near 0, while an input of approximately 5 V corresponds to a value near 1023.

The analogRead() function is used to acquire the sensor value: **sensorValue = analogRead(A0);**

The sensor value can then be converted into voltage using: **voltage = sensorValue × 5.0 / 1023.0;**

The Serial.begin(9600) command initializes serial communication at a baud rate of 9600 bits per second. The Serial.print() and Serial.println() functions transmit the measured sensor value and calculated voltage to the computer.

By repeatedly reading the sensor and transmitting the results, the Arduino provides **real-time data acquisition**. Changes in light intensity can therefore be observed immediately through the Arduino Serial Monitor or Serial Plotter.

Objectives

- Understand the working principle of an analog sensor and its application in robotic and automation systems.
- Understand the concept of analog signal acquisition using an Arduino UNO.
- Identify the Arduino UNO analog input terminals and understand their functions.
- Interface an LDR-based analog sensor circuit with the Arduino UNO.
- Understand the operation of a voltage-divider circuit used for sensor interfacing.
- Understand the working principle of the Arduino UNO's 10-bit ADC.
- Convert the acquired ADC value into a corresponding sensor voltage.
- Develop an Arduino program to continuously read analog sensor data.
- Observe and analyze sensor readings in real time using the Serial Monitor.
- Observe the variation of sensor readings with changes in light intensity.
- Apply software-based data acquisition and monitoring techniques.
- Relate analog sensor measurements to practical robotic and automation applications.

Components Required (Tinkercad/Wokwi and Hardware)

Component	Quantity
Arduino UNO	1
LDR (Photoresistor)	1
10 kΩ Resistor	1
Breadboard	1
Jumper Wires	As required
USB Cable / Power Source	1

Arduino Program – Basic Analog Data Acquisition

```
const int S = A0;
const int LED = 2;
void setup()
{
Serial.begin(9600); pinMode(LED, OUTPUT);
}
void loop()
{
int a = analogRead(S); float v = a * 5.0 / 1023.0;
Serial.print("ADC=");
Serial.print(a);
Serial.print(" Voltage=");
Serial.println(v, 2);
digitalWrite(LED, a < 500);
delay(200);
}
```

Program Description

The program begins by defining **A0** as the analog sensor input pin. During setup, serial communication is initialized at **9600 baud**. Inside the loop() function, the analogRead() function continuously measures the voltage generated by the LDR voltage-divider circuit. The resulting 10-bit ADC value is stored in the variable sensorValue.

The ADC value is converted into an approximate voltage using the equation:

Voltage = ADC Value × 5.0 / 1023.0

The sensor value and corresponding voltage are then transmitted to the computer through the Serial Monitor. A short delay is included between successive measurements to make the real-time readings easier to observe.

6. Software & Resources Required

Resource	Purpose
Tinkercad (www.tinkercad.com)	Primary simulation platform for circuit design, Arduino programming, and servo motor position-control activities.
Wokwi (wokwi.com)	Alternative simulation platform for Arduino-based circuit design, programming, and testing of servo motor position control.
Web Browser (Chrome/Edge, updated)	Tinkercad and Wokwi run directly in the browser; no software installation is required.
Projector / Screen Share	Enables live instructor demonstrations of circuit wiring, programming, simulation, and servo operation.
Observation Sheet	Used by students to record potentiometer positions, ADC values, servo angles, observations, and experimental outcomes.

*This experiment follows a **Hardware-Integrated and Simulation-Based** approach. Physical components, kits, and Arduino-based hardware are used for hands-on implementation and verification, while **Tinkercad /Wokwi** are used for circuit design, programming, simulation, and servo motor position-control testing.*

7. Student Preparation (Pre-Lab)

- Create a free Tinkercad/Wokwi account prior to the session or join the instructor's Tinkercad/Wokwi Classroom using the provided class code.
- Review the pre-lab material on analog sensors, voltage dividers, Arduino UNO, ADC, analog signal interfacing, and real-time data acquisition before the session.
- Bring a notebook to record circuit connections, sensor readings, ADC values, real-time observations, and experimental results.

8. Minute-wise Instructor & Student Activity Plan

Time	Instructor Activity	Student Activity	Domain
0–30 min	Introduce the experiment, objectives, and applications of analog sensors and real-time data acquisition in embedded and robotic systems.	Listen and understand the experiment objectives, sensor applications, and real-time data acquisition concept.	Affective / Cognitive
30–65 min	Explain the working principle of analog sensors, sensor output characteristics, and the difference between analog and digital signals.	Observe sensor characteristics and identify the role of analog signals in measurement and control applications.	Cognitive
65–100 min	Explain Arduino analog input operation, ADC resolution, reference voltage, and the relationship between sensor voltage and ADC value.	Observe analog input operation and relate sensor output voltage to corresponding ADC readings.	Cognitive
100–140 min	Demonstrate the Arduino UNO, analog sensor, VCC, GND, and analog input connections in Tinkercad/Wokwi.	Build and verify the simulated analog sensor interfacing circuit.	Psychomotor
140–185 min	Explain ADC operation and demonstrate conversion of analog sensor voltage into corresponding digital values (0–1023).	Observe ADC values for different sensor conditions and record the corresponding readings.	Cognitive
185–230 min	Explain and demonstrate the Arduino program using analogRead() and Serial.begin() / Serial.println() for real-time data monitoring.	Run the program and identify the function of each major statement used for data acquisition and serial monitoring.	Cognitive / Psychomotor
230–275 min	Guide students in varying the sensor input and observing the real-time sensor-value relationship through the Serial Monitor/Serial Plotter.	Operate the sensor and record at least five observations of sensor input and corresponding ADC values.	Psychomotor
275–315 min	Demonstrate sensor calibration and explain scaling of ADC readings into meaningful physical units or percentage values.	Modify the program, verify the scaled output, and compare measured values with expected sensor behavior.	Psychomotor
315–345 min	Discuss troubleshooting, sensor noise, sampling variations, calibration errors, and real-time data acquisition applications in automation and robotics.	Analyze results, troubleshoot errors, and relate the experiment to practical sensing and monitoring applications.	Cognitive / Affective
345–360 min	Conduct review and Q&A, summarize the experiment, and provide post-lab instructions.	Answer questions, summarize key concepts, and note post-lab work.	Cognitive

9. Instructor Demonstration Steps (Simulation and Hardware Implementation)

- **Step-1:** Open the Arduino simulation environment or connect the physical Arduino UNO to the computer.
- **Step-2:** Place the Arduino UNO, LDR, 10 kΩ resistor, and breadboard in the workspace.
- **Step-3:** Construct the LDR voltage-divider circuit according to the specified wiring connections.
- **Step-4:** Connect the voltage-divider output to analog input A0.
- **Step-5:** Connect the circuit supply to 5 V and GND of the Arduino UNO.
- **Step-6:** Enter or upload the Arduino program.
- **Step-7:** Start the simulation or power the physical Arduino.
- **Step-8:** Open the Serial Monitor and set the baud rate to 9600 baud.
- **Step-9:** Observe the sensor value and calculated voltage.
- **Step-10:** Expose the LDR to different levels of light and observe the change in ADC readings.
- **Step-11:** Cover the LDR partially or completely and observe the corresponding sensor response.
- **Step-12:** Record the sensor readings for different lighting conditions.
- **Step-13:** Open the Serial Plotter, if available, to observe the sensor response graphically in real time.
- **Step-14:** Compare the changes in light intensity with the acquired sensor values and voltage.
- **Step-15:** Stop the simulation, verify the circuit and program, save the project, and rename it.

10. Guided / Probing Questions

- What is meant by analog sensor interfacing and real-time data acquisition?
- What is the purpose of the analog sensor in this experiment?
- Why is the sensor output connected to an Arduino analog input pin such as A0?
- What is the ADC resolution and range used by the Arduino UNO in this experiment?
- What does the analog Read() function do in the sensor data acquisition program?
- Why is the Arduino UNO suitable for acquiring analog sensor data?
- What is the purpose of Serial.begin() in the Arduino program?
- What happens to the ADC reading when the sensor output voltage increases?
- Why should the Arduino and analog sensor share a common ground?
- For an ADC value near 512, what sensor voltage is approximately obtained when the Arduino reference is 5 V?
- How would you modify the program to convert the ADC reading into a percentage value from 0% to 100%?
- Why is sensor calibration important in real-time data acquisition?
- What is the role of the Serial Monitor or Serial Plotter in this experiment?
- Give two possible reasons for fluctuations or differences between expected and observed sensor readings.
- Suggest one practical application in which an analog sensor can be used for real-time monitoring or control in an embedded or robotic system.

11. Hands-on Activity (Student Task)

Task: Build and simulate an analog sensor interfacing and real-time data acquisition circuit using **Tinkercad / Wokwi**.

- Log in and create a project named **Analog_Sensor_Data_Acquisition_[YourName]_[RollNo]**.

- Place an **Arduino UNO, analog sensor, breadboard, and jumper wires**, and complete the required connections.
- Run the Arduino program to read the analog sensor output through an analog input pin and display the corresponding ADC values in real time.
- Record at least five observations of sensor input/output conditions and corresponding ADC values, and verify the sensor response and scaled readings.
- Modify the program, if required, to convert ADC readings into voltage or percentage values and observe the real-time changes.
- Save the project and keep the **Tinkercad/Wokwi project link** ready for instructor review.

12. Discussion Points

- **Analog sensor applications in embedded and robotic systems** - Discuss the use of analog sensors for temperature, light, pressure, position, force, and other real-time measurements.
- **Analog data acquisition concept** - Discuss how the analog sensor generates a varying voltage, which the **Arduino UNO** converts through its ADC into a digital value for processing and monitoring.
- **ADC, calibration, and real-time monitoring** - Discuss ADC resolution, sensor calibration, conversion of ADC values into voltage or meaningful units, and the use of the Serial Monitor/Serial Plotter for observing real-time data.
- **Simulation, hardware integration, and reliable operation** - Discuss the use of **Tinkercad/Wokwi** for simulation and testing, along with proper wiring, common grounding, systematic observations, sensor calibration, and organized project management.

13. Reflection Questions (End of Session)

- What is one new concept you learned about analog sensor interfacing and real-time data acquisition today?
- Which part of the Tinkercad/Wokwi simulation did you find most intuitive, and which part was confusing?
- How confident do you feel about explaining the relationship between sensor output voltage, ADC value, and the corresponding real-time sensor reading (rate 1–5), and why?
- What did you learn about converting ADC readings into voltage or percentage values?
- How confident are you in identifying possible causes of fluctuating or unexpected sensor readings?

14. Home Assignment

- Modify the Tinkercad/Wokwi analog sensor circuit and record five sets of sensor conditions, ADC values, and corresponding sensor output voltage.
- Modify the Arduino program to convert the ADC readings into voltage or percentage values and verify the operation through simulation.
- Explore the effect of changing the sensor input and record the corresponding real-time ADC readings using the Serial Monitor/Serial Plotter.
- Write a short note explaining one practical application of analog sensor-based real-time data acquisition, such as temperature monitoring, light sensing, pressure measurement, or robotic sensing.

15. Assessment Rubrics

Rating Scale: 4 = Excellent, 3 = Good, 2 = Satisfactory, 1 = Needs Improvement (aligned to course domain weightage: Cognitive 40% / Psychomotor 45% / Affective 15%)

Criterion	Domain	4 – Excellent	3 – Good	2 – Satisfactory	1 – Needs Improvement
Analog Sensor Identification & Circuit Construction	Cognitive / Psychomotor	Correctly identifies the analog sensor and independently constructs the interfacing circuit with accurate connections.	Identifies the sensor and builds the circuit with minor errors.	Identifies the sensor and requires assistance with circuit construction.	Unable to identify the sensor or construct the circuit correctly.
Sensor Interfacing & ADC Measurement	Cognitive / Psychomotor	Successfully interfaces the analog sensor, reads ADC values accurately, and clearly explains the relationship between sensor output and ADC readings.	Successfully interfaces the sensor and records most ADC readings correctly.	Performs sensor interfacing with assistance and records limited observations.	Unable to interface the sensor or obtain ADC readings correctly.
Real-Time Data Acquisition & Monitoring	Cognitive / Psychomotor	Correctly implements real-time data acquisition, displays sensor values accurately, and independently troubleshoots errors.	Runs the data-acquisition program and displays sensor values with minor errors or assistance.	Requires considerable assistance with programming or real-time data acquisition.	Unable to run or modify the program or acquire real-time sensor data.
Laboratory Engagement & Teamwork	Affective	Participates actively, follows laboratory practices, maintains an organized workspace, and supports peers.	Participates regularly and cooperates well with peers.	Participates when prompted and shows limited teamwork.	Remains passive, disengaged, or does not cooperate with peers.

16. Student Observation Sheet (Template)

Sl. No.	Condition	ADC Value (0–1023)	Approx. Voltage (V)
1	Bright light	Record value	Record value
2	Normal room light	Record value	Record value
3	Low light	Record value	Record value
4	LDR partially covered	Record value	Record value
5	LDR fully covered	Record value	Record value

17. Expected Learning Outcome

At the end of the session, a student should be able to independently open Tinkercad/Wokwi, construct and simulate an analog sensor interfacing circuit using Arduino UNO, correctly program the ADC-based sensor data acquisition, monitor and record real-time sensor values, analyze the relationship between sensor output and ADC readings, and explain in their own words how analog sensors are interfaced and used for real-time data acquisition in embedded and robotic applications.

18. Instructor Tips

- Keep the initial explanation concise and focused on analog sensor operation, Arduino UNO, ADC, sensor interfacing, and real-time data acquisition.
- Closely monitor students during the hands-on activity, particularly while making Arduino–analog sensor connections and troubleshooting wiring or programming errors.
- Encourage peer-assisted learning and emphasize the relationship between sensor stimulus → analog sensor output → ADC value → real-time data display.
- Maintain the Hardware-Integrated and Simulation-Based approach, using physical hardware for hands-on verification and Tinkercad/Wokwi for circuit design, programming, simulation, and testing.

CSE 4571 — Practical Robotics Project with Arduino

Basic Questions

No.	Question	Learning Domain	Taxonomy Level
1	What is an analog sensor?	Cognitive	Bloom's Taxonomy – Remember
2	Name any two applications of analog sensors in robotics.	Cognitive	Bloom's Taxonomy – Remember
3	What is meant by analog sensor interfacing?	Cognitive	Bloom's Taxonomy – Understand
4	What is the role of Arduino UNO in analog sensor data acquisition?	Cognitive	Bloom's Taxonomy – Understand
5	What is an ADC?	Cognitive	Bloom's Taxonomy – Remember
6	Why is an ADC required when interfacing an analog sensor with Arduino UNO?	Cognitive	Bloom's Taxonomy – Understand
7	What is the ADC resolution of the Arduino UNO?	Cognitive	Bloom's Taxonomy – Remember
8	What is the typical ADC input range of Arduino UNO?	Cognitive	Bloom's Taxonomy – Remember
9	Which Arduino function is commonly used to read an analog sensor value?	Cognitive	Bloom's Taxonomy – Remember
10	Which Arduino pin is used to connect an analog sensor output?	Cognitive	Bloom's Taxonomy – Remember
11	What is the relationship between sensor output voltage and ADC value?	Cognitive	Bloom's Taxonomy – Understand
12	What happens to the ADC reading when the analog sensor output voltage increases?	Cognitive	Bloom's Taxonomy – Analyze

13	Why should the sensor and Arduino share a common ground?	Cognitive	Bloom's Taxonomy – Understand
14	How is an analog sensor signal converted into a digital value by Arduino UNO?	Cognitive	Bloom's Taxonomy – Understand
15	Calculate the approximate ADC value for an input voltage of 2.5 V using a 10-bit ADC with a 5 V reference.	Cognitive	Bloom's Taxonomy – Apply
16	Calculate the approximate sensor voltage corresponding to an ADC value of 512 using a 5 V reference.	Cognitive	Bloom's Taxonomy – Apply
17	What ADC value corresponds approximately to the maximum input voltage of 5 V?	Cognitive	Bloom's Taxonomy – Apply
18	Why is real-time data monitoring useful in sensor-based robotic systems?	Cognitive	Bloom's Taxonomy – Understand
19	Can you locate the Circuits option in Tinkercad or create a new Arduino project in Wokwi?	Psychomotor	Simpson's Taxonomy – Perception
20	Can you identify the Arduino UNO, analog sensor, jumper wires, and other required components in the workspace?	Psychomotor	Simpson's Taxonomy – Perception
21	Can you place the Arduino UNO and analog sensor correctly on the virtual workspace?	Psychomotor	Simpson's Taxonomy – Set
22	Can you connect the analog sensor to 5V, GND, and an analog input pin correctly?	Psychomotor	Simpson's Taxonomy – Guided Response
23	Can you write or modify an Arduino program to read the analog sensor using analogRead()?	Psychomotor	Simpson's Taxonomy – Guided Response
24	Can you start the simulation and verify that the Arduino reads the sensor value correctly?	Psychomotor	Simpson's Taxonomy – Mechanism
25	Can you observe the real-time sensor readings through the Serial Monitor or Serial Plotter?	Psychomotor	Simpson's Taxonomy – Mechanism
26	Can you vary the sensor input and record the corresponding ADC values and sensor readings?	Psychomotor	Simpson's Taxonomy – Mechanism
27	Can you troubleshoot incorrect sensor readings by checking the circuit connections and program?	Psychomotor	Simpson's Taxonomy – Complex Overt Response
28	Are you willing to follow proper simulation-laboratory practices while working with Tinkercad/Wokwi?	Affective	Krathwohl's Taxonomy – Receiving
29	How actively did you participate and support your peers during the analog sensor interfacing and real-time data acquisition activity?	Affective	Krathwohl's Taxonomy – Responding
30	Do you value real-time analog sensor data acquisition as an effective method for monitoring and controlling robotic systems? Why?	Affective	Krathwohl's Taxonomy – Valuing